



School of Computer Sciences and IT
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Ride Sharing System

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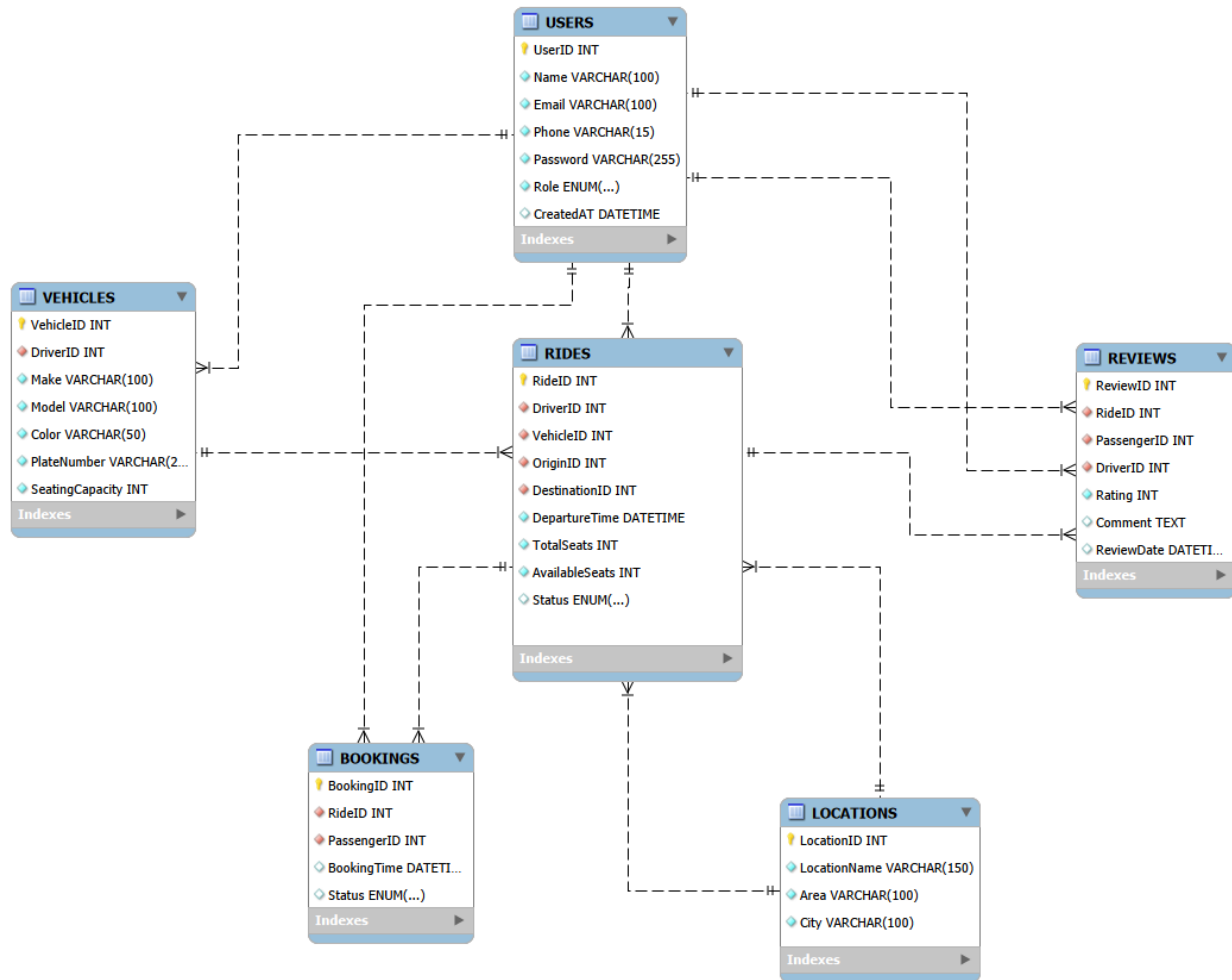
Submitted To: Sir Ali Hassan

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Project Milestones

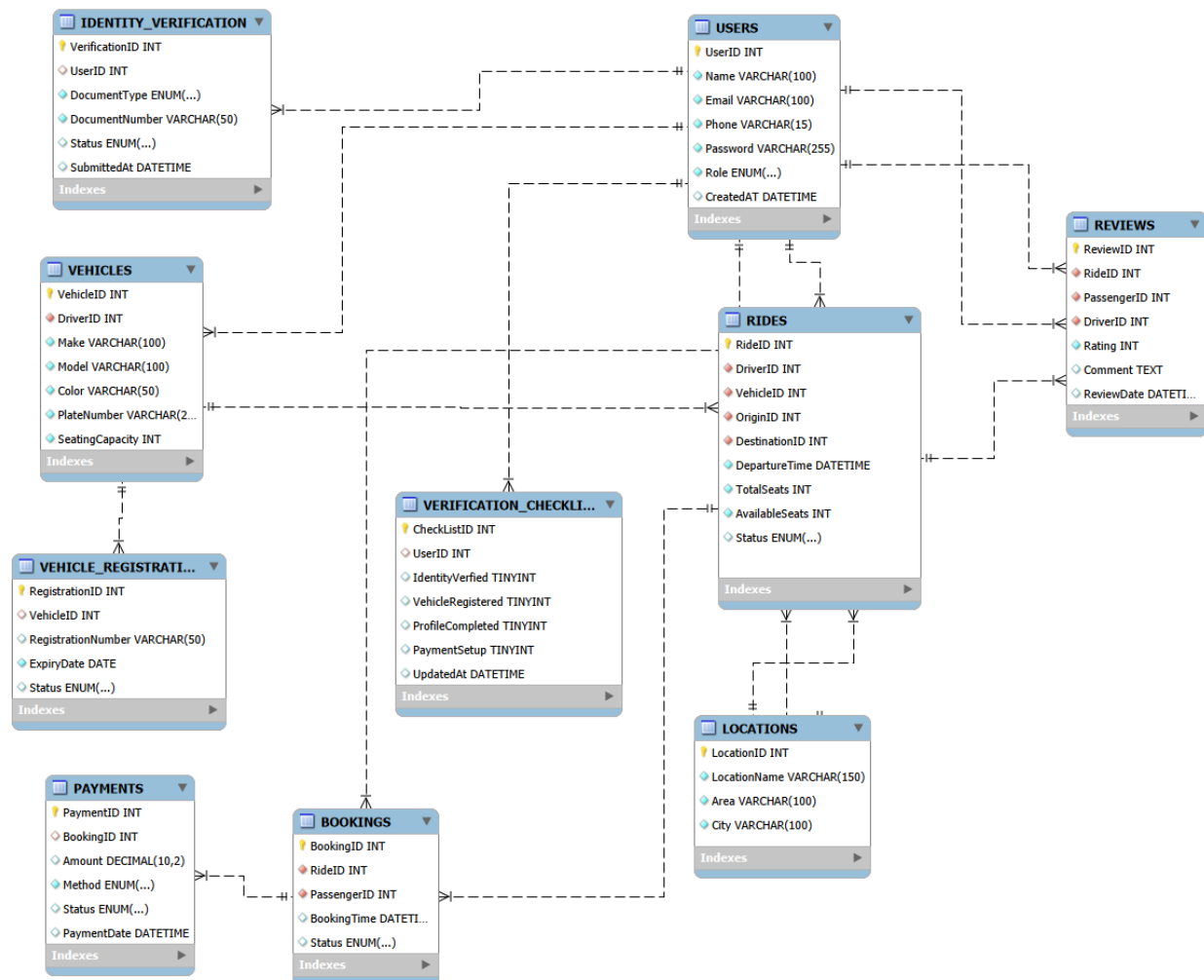
Milestone	Version	Date	Remarks
<i>Milestone 1</i> – ERD & Relational Schema	V1.0	26/04/2026	
<i>Milestone 1</i> – Schema Update	V1.1	11/05/2026	

1. Entity-Relationship Diagram (ERD) (V1.0)



Entity-Relationship Diagram – Ride Sharing System

1. Entity-Relationship Diagram (ERD) (V1.1)



2. Relational Database Schema

This section presents the relational schema derived from the Entity-Relationship Diagram (ERD). Each entity is transformed into a table with clearly defined primary keys (PK) and foreign keys (FK) to maintain relationships and data integrity.

2.1 USERS

Description: Stores all system users, including both drivers and passengers.

USERS(UserID PK, Name, Email, Phone, Password, Role, CreatedAt)

	Field	Type	Null	Key	Default	Extra
►	UserID	int	NO	PRI	NULL	auto_increment
	Name	varchar(100)	NO		NULL	
	Email	varchar(100)	NO	UNI	NULL	
	Phone	varchar(15)	NO		NULL	
	Password	varchar(255)	NO		NULL	
	Role	enum('Driver','Passenger')	NO		NULL	
	CreatedAt	datetime	NO		NULL	

USERS Table

2.2 VEHICLES

Description: Contains vehicle details associated with drivers.

VEHICLES(VehicleID PK, DriverID FK -> USERS(UserID), Make, Model, Color, PlateNumber, SeatingCapacity)

	Field	Type	Null	Key	Default	Extra
►	VehicleID	int	NO	PRI	NULL	auto_increment
	DriverID	int	NO	MUL	NULL	
	Make	varchar(100)	NO		NULL	
	Model	varchar(100)	NO		NULL	
	Color	varchar(50)	NO		NULL	
	PlateNumber	varchar(20)	NO		NULL	
	SeatingCapacity	int	NO		NULL	

VEHICLES Table

2.3 LOCATIONS

Description: Stores pickup and destination location details.

LOCATIONS(LocationID PK, LocationName, Area, City)

	Field	Type	Null	Key	Default	Extra
►	LocationID	int	NO	PRI	NULL	auto_increment
	LocationName	varchar(150)	NO		NULL	
	Area	varchar(100)	NO		NULL	
	City	varchar(100)	NO		NULL	

LOCATIONS Table

2.4 RIDES

Description: Represents rides created by drivers, including route, timing, and seat availability.

RIDES(RideID PK, DriverID FK -> USERS(UserID), VehicleID FK -> VEHICLES(VehicleID), OriginID FK -> LOCATIONS(LocationID), DestinationID FK -> LOCATIONS(LocationID),DepartureTime, TotalSeats, AvailableSeats, Status)

	Field	Type	Null	Key	Default	Extra
►	RideID	int	NO	PRI	NULL	auto_increment
	DriverID	int	NO	MUL	NULL	
	VehicleID	int	NO	MUL	NULL	
	OriginID	int	NO	MUL	NULL	
	DestinationID	int	NO	MUL	NULL	
	RideTime	date	NO		NULL	
	DepartureTime	time	NO		NULL	
	TotalSeats	int	NO		NULL	
	AvailableSeats	int	NO		NULL	
	Status	enum('Scheduled','Ongoing','Completed')	NO		NULL	

RIDES Table

2.5 BOOKINGS

Description: Stores ride reservations made by passengers.

BOOKINGS(BookingID PK, RideID FK -> RIDES(RideID), PassengerID FK -> USERS(UserID), BookingTime, Status)

	Field	Type	Null	Key	Default	Extra
►	BookingID	int	NO	PRI	NULL	auto_increment
	RideID	int	NO	MUL	NULL	
	PassengerID	int	NO	MUL	NULL	
	BookingTime	datetime	NO		NULL	
	Status	enum('Pending','Confirmed','Cancelled')	NO		NULL	

BOOKINGS Table

2.6 REVIEWS

Description: Stores feedback and ratings given by passengers to drivers after rides.

REVIEWS(ReviewID PK, RideID FK -> RIDES(RideID), PassengerID FK -> USERS(UserID), DriverID FK -> USERS(UserID), Rating, Comment, ReviewDate)

	Field	Type	Null	Key	Default	Extra
►	ReviewID	int	NO	PRI	NULL	auto_increment
	RideID	int	NO	MUL	NULL	
	PassengerID	int	NO	MUL	NULL	
	DriverID	int	NO	MUL	NULL	
	Rating	int	NO		NULL	
	Comment	text	YES		NULL	
	ReviewDate	datetime	NO		NULL	

REVIEWS Table

2.7 PAYMENTS

Description: Stores payment transactions made by passengers for their ride bookings.

PAYMENTS(PaymentID PK, BookingID FK -> BOOKINGS(BookingID), Amount, Method, Status, PaymentDate)

	Field	Type	Null	Key	Default	Extra
►	PaymentID	int	NO	PRI	NULL	auto_increment
	BookingID	int	NO	MUL	NULL	
	Amount	decimal(10,2)	NO		NULL	
	Method	enum('cash','online')	NO		NULL	
	Status	enum('pending','completed','failed')	YES		pending	
	PaymentDate	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

2.8 IDENTITY_VERIFICATION

Description: Stores identity document submissions by users for verification purposes.

IDENTITY_VERIFICATION(VerificationID PK, UserID FK -> USERS(UserID),
DocumentType, DocumentNumber, Status, SubmittedAt)

Field	Type	Null	Key	Default	Extra
VerificationID	int	NO	PRI	NULL	auto_increment
UserID	UserID	NO	MUL	NULL	
DocumentType	enum('CNIC','student_card','passport')	NO		NULL	
DocumentNumber	varchar(50)	NO	UNI	NULL	
Status	enum('pending','verified','rejected')	YES		pending	
SubmittedAt	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

2.9 IDENTITY_VERIFICATION

Description: Stores vehicle registration details linked to each registered vehicle.

VEHICLE_REGISTRATION(RegistrationID PK, VehicleID FK -> VEHICLES(VehicleID),
RegistrationNumber, ExpiryDate, Status)

Field	Type	Null	Key	Default	Extra
RegistrationID	int	NO	PRI	NULL	auto_increment
VehicleID	int	NO	MUL	NULL	
RegistrationNumber	varchar(50)	NO	UNI	NULL	
ExpiryDate	date	NO		NULL	
Status	enum('active','expired','suspended')	YES		active	

2.10 VERIFICATION_CHECKLIST

Description: Tracks the verification progress of each user across multiple checklist items.

VERIFICATION_CHECKLIST(CheckListID PK, UserID FK -> USERS(UserID),
IdentityVerified, VehicleRegistered, ProfileCompleted, PaymentSetup,
UpdatedAt)

	Field	Type	Null	Key	Default	Extra
►	CheckListID	int	NO	PRI	HULL	auto_increment
	UserID	int	NO	MUL	HULL	
	IdentityVerified	tinyint	YES		0	
	VehicleRegistered	tinyint	YES		0	
	ProfileCompleted	tinyint	YES		0	
	PaymentSetup	tinyint	YES		0	
	UpdatedAt	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED